Supplementary Material: Simulation Parameters

Hypergame-Aware AI Defense for IoT Perception Attacks Smart Hospital Simulation

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All parameters are fixed across all 2500 simulation runs (50 seeds $\times$ 50 episodes). Randomness is controlled via `numpy.random.Generator` seeded from BASE_SEED$= 42+$ seed index. Full source code and reproducibility artifacts are available at the repository referenced in the paper.

I. Network and Simulation Configuration

Parameter	Symbol	Value	Description / Equation
Bedside monitors	N_{mon}	100	Device type M1
Infusion pumps	N_{pmp}	30	Device type M2
Ventilators	N_{ven}	20	Device type M3
Wearable sensors	N_{wear}	200	Device type M4
Total devices	N	350	Three-tier topology
Patients	N_{pat}	50	7 devices/patient avg.
State vector dim.	d	100	-
Sensor readings	n_s	5	Per device
Episodes	T_{ep}	50	Training episodes
Steps/episode	T	200	Time steps per episode
Seeds	N_{seed}	50	2500 total runs
Base seed	-	42	Reproducibility
Gateway mesh prob.	P_{gw}	0.80	Pump $\leftrightarrow$ vent, same ward
Uplink prob.	P_{up}	0.60	Monitor $\leftrightarrow$ pump/vent, same ward
Peer prob.	P_{peer}	0.20	Monitor $\leftrightarrow$ monitor, same ward

II. Attack Model Parameters

Parameter	Symbol	Value	Description / Equation
Perturbation budget	ε	0.15	ℓ_2-norm bound, Eq. 4
Sensor noise std.	σ	0.05	Gaussian noise
Base attack prob.	P_{base}	0.30	Baseline probability of attack success under neutral conditions
Multi-target penalty	λ_c	0.10	Eq. 3
Health degradation	κ	0.90	Sec. III
Trust decay rate	β_d	0.01	Sec. III
Trust recovery rate	β_r	0.05	Sec. III
Low-trust threshold	τ	0.40	Sec. III
Atk. type multipliers	μ_a	A1: 0.9 A2: 1.2 A3: 1.0 A4: 0.8 A5: 0.7	Eq.60

III. Defense Action Space

Effectiveness values η are ordered by attack type: A1 (spoofing) / A2 (adversarial) / A3 (poisoning) / A4 (extraction) / A5 (replay).

Action	Idx	Effectiveness η (A1/A2/A3/A4/A5)	Cost c_d	Notes
Monitor	D0	0.5 / 0.3 / 0.3 / 0.3 / 0.3	0.10	Baseline surveillance
Firewall	D1	1.5 / 0.5 / 0.5 / 0.5 / 0.5	0.50	vs. spoofing
Encryption	D2	0.5 / 0.5 / 1.5 / 1.5 / 0.5	1.00	vs. poisoning/extract
Honeypot	D3	1.5 / 0.5 / 0.5 / 0.5 / 1.5	0.80	vs. spoofing/replay
Deception	D4	0.5 / 1.5 / 0.5 / 0.5 / 0.5	0.60	vs. adversarial
Isolation	D5	0.5 / 0.5 / 1.5 / 0.5 / 0.5	1.20	vs. poisoning
Access Control	D6	0.8 / 0.8 / 0.8 / 0.8 / 0.8	0.30	General (capped 0.8)

Device Type	Criticality c_i	Clinical Justification
Bedside monitor	0.8	Vital sign surveillance; high but not airway-critical
Infusion pump	1.0	Wrong infusion rate is immediately life-threatening
Ventilator	1.0	Airway management; failure is fatal within minutes
Wearable sensor	0.5	Supplementary telemetry; lower immediate criticality

IV. Device Criticality Weights

V. Patient Physiological Parameters

All 50 patients are initialised from $\mathcal{N}(\mu_{\text{normal}}, \sigma)$ clipped to [min, max]. Dynamics follow AR(1): $v_{t+1} = 0.98\, v_t + 0.02\, \mu_{\text{normal}} + \varepsilon_t$, $\varepsilon_t \sim \mathcal{N}(0,\ 0.01 \cdot \sigma)$.

Vital Sign	Unit	μ_{normal}	σ	Min	Max
Heart Rate (HR)	bpm	75	10	40	200
Systolic BP (SBP)	mmHg	120	10	60	220
Diastolic BP (DBP)	mmHg	80	8	40	130
O_2 Sat. (SpO_2)	%	98	1.5	70	100
Respiratory Rate (RR)	/min	15	2	4	50
Temperature (Temp)	°C	37.0	0.3	34.0	42.0
Blood Glucose (Gluc)	mg/dL	100	15	20	600
Infusion Rate	mL/hr	50	10	0	500
Tidal Volume (TidalV)	mL	450	50	100	900
FiO_2	-	0.40	0.05	0.21	1.0
PEEP	cmH_2O	8	1.5	0	25

VI. Particle Filter Parameters

Parameter	Symbol	Value	Description
Particles per device	N	100	SIR particle count
Min eff. sample size	$N_{\text{eff}}^{\min}$	20	$0.2 \times N$ threshold
Transition noise std.	τ	0.05	AR(1) state noise
Observation noise	σ	0.05	Gaussian likelihood
Adaptive variance scale	-	$9\sigma^2$	When innovation $> 3\sigma\sqrt{d}$
Innovation window	W	10	Deception detection buffer
Deception threshold	-	$3\sigma\sqrt{d}$	Triggers variance inflation
Isolation delta scale	-	0.50	$\delta \leftarrow 0.5\,\delta$ under D5
Deception jitter std.	-	0.01	Added to δ under D4

Parameter	Symbol	Value	Description / Equation
Learning rate (Adam)	η	0.001	Outer loop, Eq. 57
Adam β_1	β_1	0.9	1st moment decay
Adam β_2	β_2	0.999	2nd moment decay
Softmax temperature	β_{temp}	0.5	IBR best response
KL alignment weight	ω	0.3	Outer loop, Eq. 57
KL scale clip max	$\gamma_{\max}$	1.3	Gradient stability
Strategy reg. weight	μ	0.05	Eq. 58 Term III
Perception adv. weight	β	0.5	Inner loop, Eq. 54
Adv. perturbation steps	n_{inner}	5	PGA inner loop
Adv. step size	α	0.01	PGA step size
KL histogram bins	B	15	P^D and P^A discretisation
MLP hidden units	-	64	2-layer ReLU network
Perception weight (game)	λ	0.10	Eq. 29

Parameter	Symbol	Value	Description
Defender reasoning level	k_D	2	k-level IBR depth
Attacker reasoning level	k_A	2	k-level IBR depth
IBR convergence threshold	$\varepsilon_{\text{conv}}$	0.001	Strategy change threshold
IBR max iterations	T_{IBR}	20	Per time step
Discount factor	γ	0.95	POMDP value function
Entropy reg. (strategy)	μ	0.05	Applied after IBR
Defender perception noise	ϕ_D	0.10	Sec. IV-E
Attacker perception noise	ϕ_A	0.20	Higher uncertainty

VII. Adversarial Meta-Learner Parameters

VIII. Hypergame Equilibrium Parameters